



Mass flow controller (MFC)/mass flow meter (MFM) for gases

- Nominal flow range from 0.01 l_N/min to 100 l_N/min (ref. nitrogen)
- High accuracy in measurement and repeatability
- Suitable for aggressive gases
- User-friendly gas conversion
- Easy device exchange due to configuration memory

Product variants described in the data sheet may differ from the product presentation and description.

Can be combined with



Type 7011

Direct-acting 2/2-way plunger valve



Type 6013

Plunger valve 2/2-way direct-acting



Type 0330

Direct-acting 2/2 or 3/2-way pivoted armature valve



Type description

The Type 8743 mass flow controller (MFC)/mass flow meter (MFM) is suitable for controlling or measuring aggressive gases, gas mixtures and in the area of application for alternating gases. A thermal capillary sensor, with non-wetted sensor elements, is used for this. This also enables easy conversion to the process gas in question. The wetted parts are made of high-quality stainless steel and FFKM as seal material for high chemical resistance. The MFC variant achieves high control accuracy thanks to the use of Bürkert proportional valve technology and the application-appropriate layout of the valve orifice. The MFC can be implemented with low pressure loss as required, due to the diversity of the available valve orifices. These MFCs and MFMs either communicate the set-point and actual values of the flow value via an analogue interface, or other values, as well as the set-point and actual values, that can be selected via software via an Ethernet interface. Type 8744 represents the IP65 variant of the Type 8743 and contains the CAN-based bÜS variant for integration into CAN or bÜS networks. A wide range of devices can easily and efficiently be integrated into the control level in this way via an Ethernet gateway. The communication data is configured via the Bürkert Communicator software.

Table of contents

1. General technical data	3
2. Approvals and conformities	4
2.1. General notes	4
2.2. Conformity	4
2.3. Standards	4
2.4. North America (USA/Canada)	4
2.5. Foods and beverages/Hygiene	5
2.6. Oxygen	5
3. Materials	5
3.1. Bürkert resistApp	5
4. Dimensions	6
4.1. Threaded variant connections	6
Small nominal flow rates < 20 l _N /min, with internal thread	6
Small nominal flow rates < 20 l _N /min, compatible with VCR®	6
Small nominal flow rates < 20 l _N /min, with clamp connection DN 15...1/2"	6
Large nominal flow rates > 20 l _N /min, with internal thread	7
4.2. Housing with interface for analogue signals, Industrial Ethernet and Modbus RTU	7
MFC with valve Type 2871 for nominal flow rates < 20 l _N /min	7
MFC with valve Type 2871 for nominal flow rates > 20 l _N /min	9
MFC with valve Type 2873 for nominal flow rates > 20 l _N /min	11
MFM for nominal flow rates < 20 l _N /min	13
MFM for nominal flow rates > 20 l _N /min	14
5. Product connections	15
5.1. Communication	15
Industrial Ethernet	15
Analogue	15
Modbus RTU	16
6. Performance specifications	17
6.1. MFM pressure loss diagram	17
7. Product operation	17
7.1. Measurement principle	17
8. Product accessories	18
8.1. Bürkert Communicator software	18
8.2. Connecting Type 8743 with the Bürkert Communicator software	19
8.3. Configuration management for easy device replacement	19
8.4. Web server for Industrial Ethernet variant	19
9. Ordering information	19
9.1. Bürkert eShop	19
9.2. Recommendation regarding product selection	19
9.3. Bürkert product filter	20
9.4. Bürkert Product Enquiry Form	20
9.5. Ordering chart accessories	20

1. General technical data

Product properties		
Dimensions	Further information can be found in chapter “4. Dimensions” on page 6.	
Material		
Seal	FKM, EPDM and FFKM, seat seal PCTFE for nominal diameters DN 0.05 and DN 0.1	
Housing	PC (polycarbonate)	
Base block	Stainless steel 1.4404/316L	
Wetted parts	Stainless steel 1.4401/316, 1.4404/316L, 1.4435/316L, PCTFE and seal material	
Total weight	Variant for nominal flow rates < 20 l _N /min: • Approx. 700 g (MFM, without valve) • Approx. 1000 g (MFC with valve Type 2871)	Variant for nominal flow rates > 20 l _N /min: • Approx. 900 g (MFM, without valve) • Approx. 1400 g (MFC with valve Type 2871) • Approx. 1500 g (MFC with valve Type 2873)
Configuration management	Further information can be found in chapter “8.3. Configuration management for easy device replacement” on page 19.	
LED display ^{1.)}	RGB LED according to NAMUR NE107	
Performance data		
Nominal flow range (Q _N) ^{2.)}	10 ml _N /min...100 l _N /min (N ₂) ^{2.)}	
Operating pressure ^{3.)}	MFM: max. 10 bar For MFCs, the maximum operating pressure depends on the medium and nominal valve size.	
Measuring accuracy ^{4.)}	± 0.8 % MV ^{5.)} ± 0.3 % FS ^{6.)} (under calibration conditions and after < 30 min warm-up time to achieve the best measuring conditions)	
Repeatability	± 0.1 % FS	
Turndown ratio	1:20 at Q _N < 25 ml/min, 1:50 at Q _N > 25 ml/min (higher on request) ^{2.)}	
Temperature coefficient	± (0.05 % FS + 0.05 % MV)/K (deviation from gas temperature during calibration and adjustment)	
Pressure coefficient	± 0.1 % MV/bar (deviation from operating pressure during calibration and adjustment, referring to N ₂)	
Settling time (MFC)/Response time (MFM) (t ₉₅)	< 2 s	
Actuator (proportional valve)		
Valve position	Normally closed	
Nominal diameter	0.05...4 mm	
K _{vs} value range	0.00006...0.32 m ³ /h	
Electrical data		
Operating voltage	24 V DC	
Power consumption ^{7.)}	MFM: 1...2 W MFC: max. 3...18 W (depending on the type of proportional valve)	
Residual ripple	± 2 %	
Voltage tolerance	± 10 %	
Electrical connection		
Analogue variant	D-Sub plug, 9-pin ^{8.)} or terminal strip, 6-pin	
Industrial Ethernet variant	2 x RJ45 socket (switch) ^{9.)}	
Modbus RTU variant	D-Sub plug, 9-pin	
Medium data		
Operating medium	Aggressive and neutral, pure gases (others on request)	
Calibration medium	Nitrogen	
Medium temperature	- 10 °C...+ 50 °C	
Product connections		
Analogue interface	4...20 mA, 0...20 mA, 0...10 V or 0...5 V Input impedance: > 20 kΩ (voltage) resp. < 300 Ω (current) Maximum current: 10 mA (voltage output) Maximum load: 400 Ω (current output)	
Digital communication interface	Modbus RTU, Industrial Ethernet: EtherCAT®, EtherNet/IP, Modbus TCP, PROFINET	
Port connection	G ¼, NPT ¼, compatible with VCR® ¼, compression fitting, clamp connection (others on request)	

Approvals and conformities

Certificate	Material certificate 3.1 (optionally)
North America (USA/Canada)	Further information can be found in chapter "2.4. North America (USA/Canada)" on page 4.
Foods and Beverages/Hygiene	Further information can be found in chapter "2.5. Foods and beverages/Hygiene" on page 5.
Oxygen	Further information can be found in chapter "2.6. Oxygen" on page 5.

Environment and installation

Installation position	Horizontal or vertical ^{10.)}
Storage temperature	- 10 °C...+ 70 °C
Relative air humidity	Max. 95 % at + 55 °C, non-condensing
Degree of protection	IP20
Ambient temperature	- 10 °C...+ 40 °C (higher temperatures on request)

Accessories

Software	Bürkert Communicator Further information can be found in chapter "8.1. Bürkert Communicator software" on page 18.
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- 1.) For a detailed description of LED colours see **operating instructions Type 8743** ►
- 2.) Index N: flow rates with respect to 1.013 bar abs and 0 °C
- 3.) Overpressure to atmospheric pressure
- 4.) If the operating medium differs from the calibration medium, the actual measuring accuracy may deviate from the specified value. If the operating medium is natural gas, the measuring accuracy depends on the composition of the natural gas, which can vary depending on origin and season.
- 5.) Of measured value
- 6.) Of full scale
- 7.) Information based on the typical power consumption (at + 23 °C ambient temperature, nominal flow rate and 30 min closed-loop control mode). The specifications according to UL 61010 - 1 may deviate (see **operating instructions Type 8743** ►).
- 8.) An additional digital input and a relay output are available for the analogue variant with D-Sub, 9-pin.
- 9.) Supply voltage via separate terminal strip
- 10.) Calibrated in horizontal installation position, Zero-point adjustment is necessary when choosing a different installation position.

2. Approvals and conformities

2.1. General notes

- The approvals and conformities listed below must be stated when making enquiries. This is the only way to ensure that the product complies with all required specifications.
- Not all available variants can be supplied with the below mentioned approvals or conformities.

2.2. Conformity

In accordance with the Declaration of Conformity, the product is compliant with the EU Directives.

2.3. Standards

The applied standards which are used to demonstrate compliance with the EU Directives are listed in the EU-Type Examination Certificate and/or the EU Declaration of Conformity.

2.4. North America (USA/Canada)

Approval	Description
	Optional: UL Listed for the USA and Canada The products are UL Listed for the USA and Canada according to: - UL 61010-1 (ELECTRICAL EQUIPMENT FOR MEASUREMENT, CONTROL, AND LABORATORY USE – Part 1: General Requirements) - CAN/CSA-C22.2 No. 61010-1

2.5. Foods and beverages/Hygiene

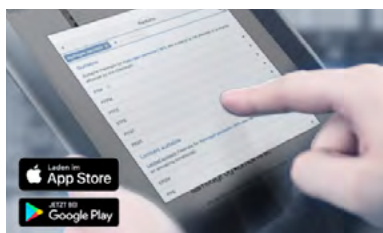
Conformity	Description
FDA	FDA – Code of Federal Regulations (valid for variable code PL02, PL03) All wetted materials are compliant with the Code of Federal Regulations published by the FDA (Food and Drug Administration, USA) according to the manufacturer's declaration.
USP	United States Pharmacopeial Convention (USP) (valid for variable code PL04) All wetted materials are biocompatible according to the manufacturer's declaration.
	EC Regulation 1935/2004 of the European Parliament and of the Council (valid for variable code PL01, PL02) All wetted materials are compliant with EC Regulation 1935/2004/EC according to the manufacturer's declaration.

2.6. Oxygen

Conformity	Description
O_2	Optional: Suitability for oxygen (valid for the variable code NL02) The products are suitable for use with gaseous oxygen, according to the manufacturer's declaration.

3. Materials

3.1. Bürkert resistApp



Bürkert resistApp – Chemical Resistance Chart

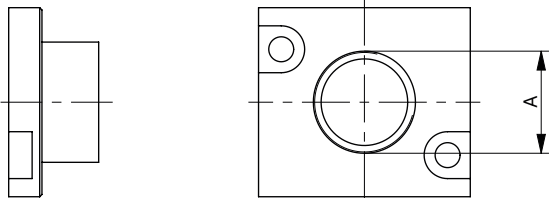
You want to ensure the reliability and durability of the materials in your individual application case? Verify your combination of media and materials on our website or in our resistApp.

[Start chemical resistance check](#)

4. Dimensions

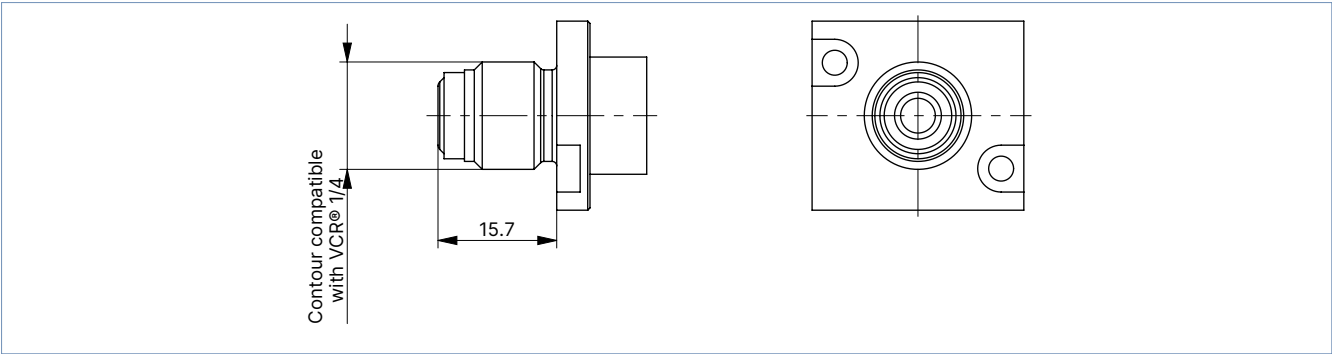
4.1. Threaded variant connections

Small nominal flow rates < 20 l_N/min, with internal thread

Dimensions	Thread (A)	Thread depth [mm]
	G 1/4	12
	NPT 1/4	12

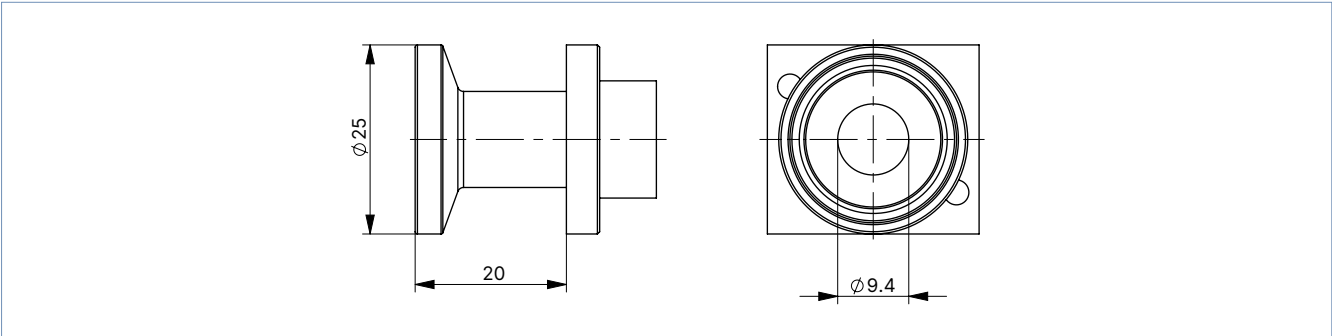
Small nominal flow rates < 20 l_N/min, compatible with VCR®

Note:
Dimensions in mm



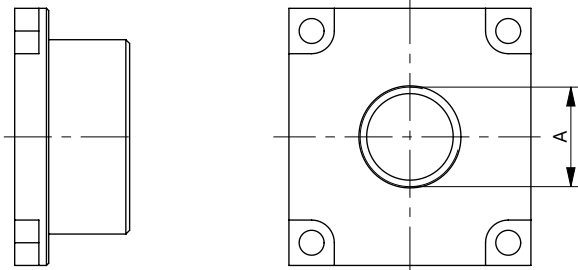
Small nominal flow rates < 20 l_N/min, with clamp connection DN 15...1/2"

Note:
Dimensions in mm



Type 8743

Large nominal flow rates > 20 l_N/min, with internal thread

Dimensions	Thread (A)	Thread depth [mm]
	G ¼	12.5
	NPT ¼	10
	G ⅜	12.5
	NPT ⅜	11
	G ½	15
	NPT ½	14

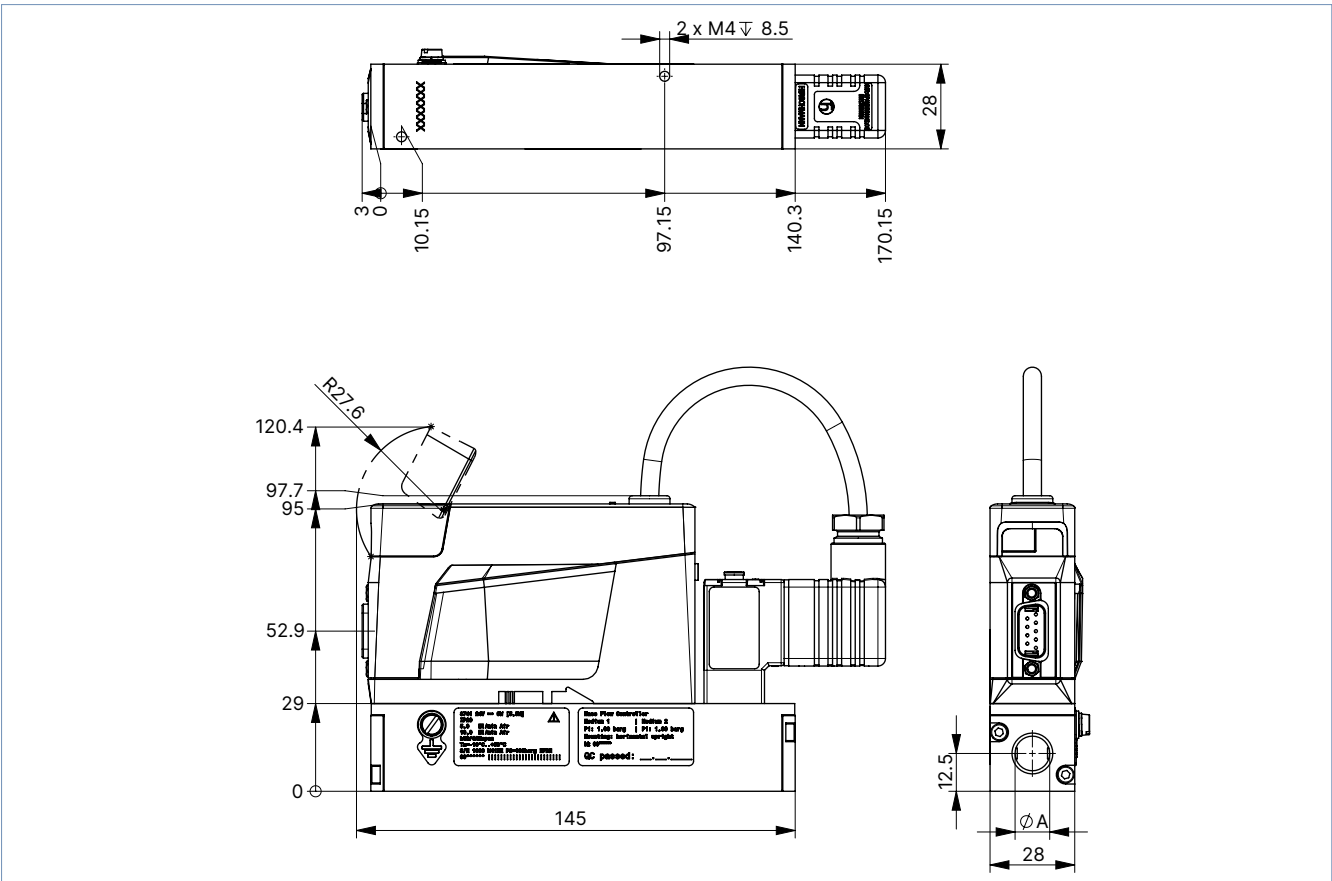
4.2. Housing with interface for analogue signals, Industrial Ethernet and Modbus RTU

MFC with valve Type 2871 for nominal flow rates < 20 l_N/min

NPT/G ¼ variant

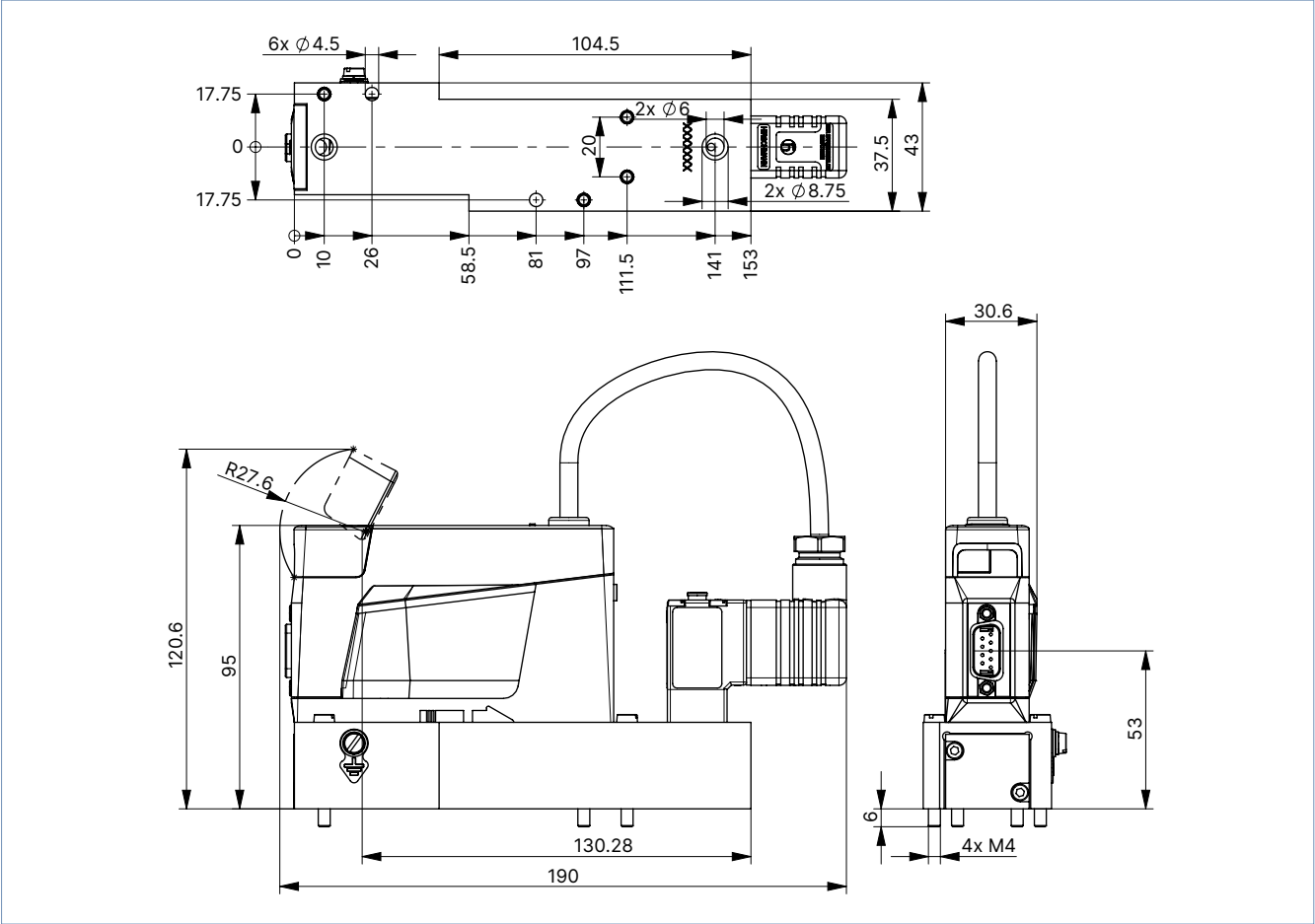
Note:

Dimensions in mm



Sub-base variant

Note:
Dimensions in mm

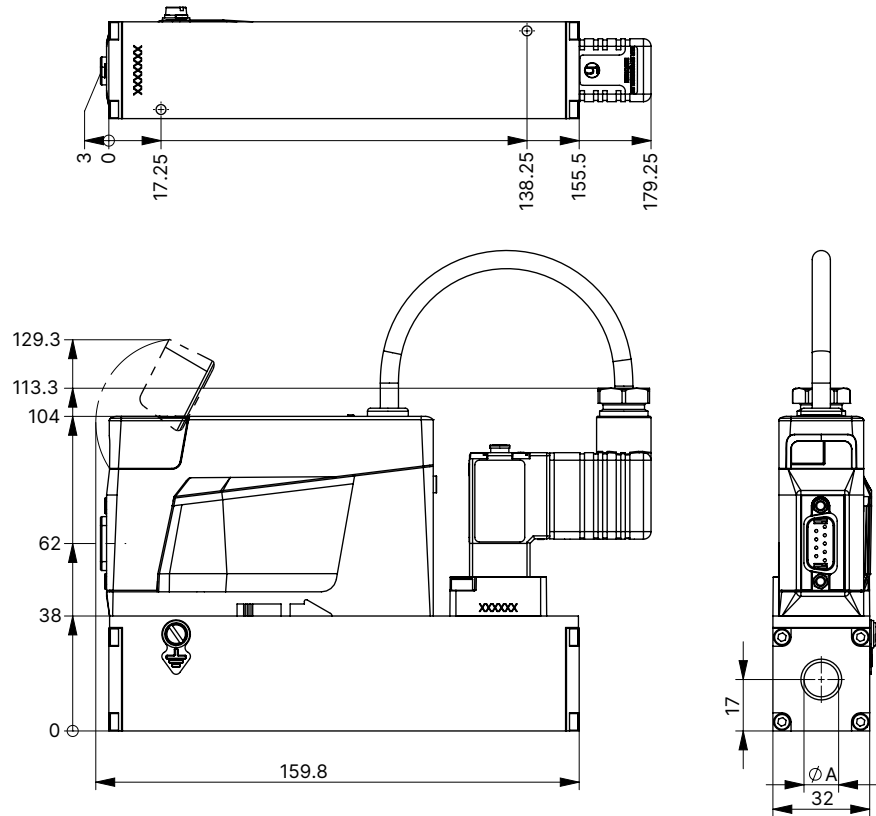


MFC with valve Type 2871 for nominal flow rates $> 20 \text{ l}_N/\text{min}$

NPT/G $\frac{1}{4}$ variant

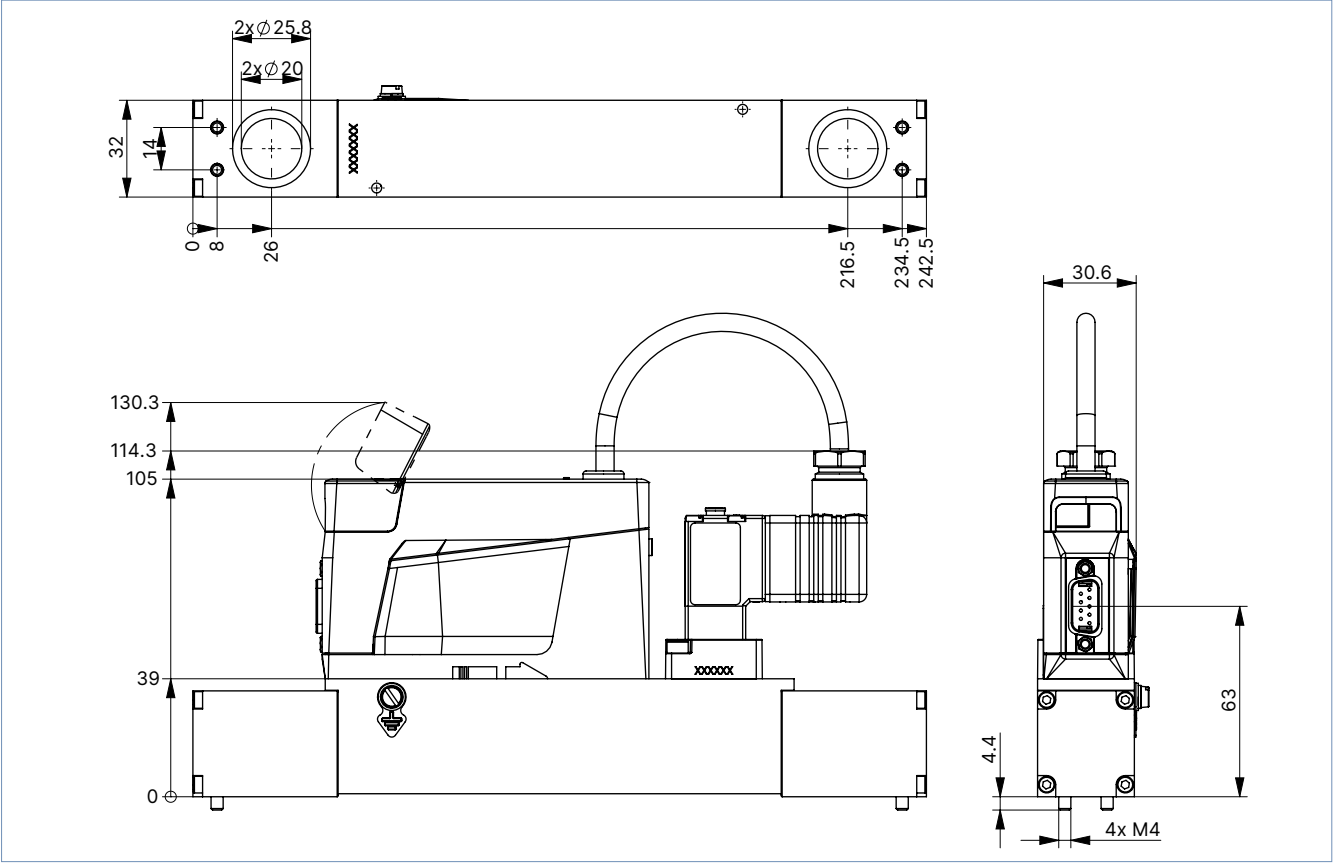
Note:

Dimensions in mm



Sub-base variant

Note:
Dimensions in mm

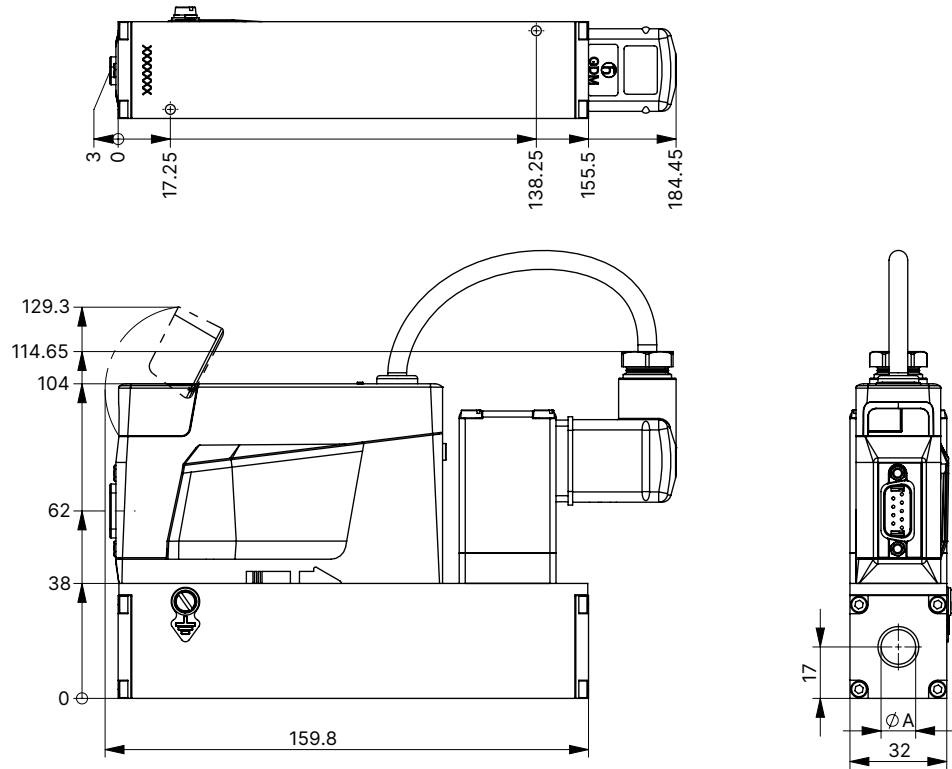


MFC with valve Type 2873 for nominal flow rates $> 20 \text{ l}_N/\text{min}$

NPT/G $\frac{1}{4}$ variant

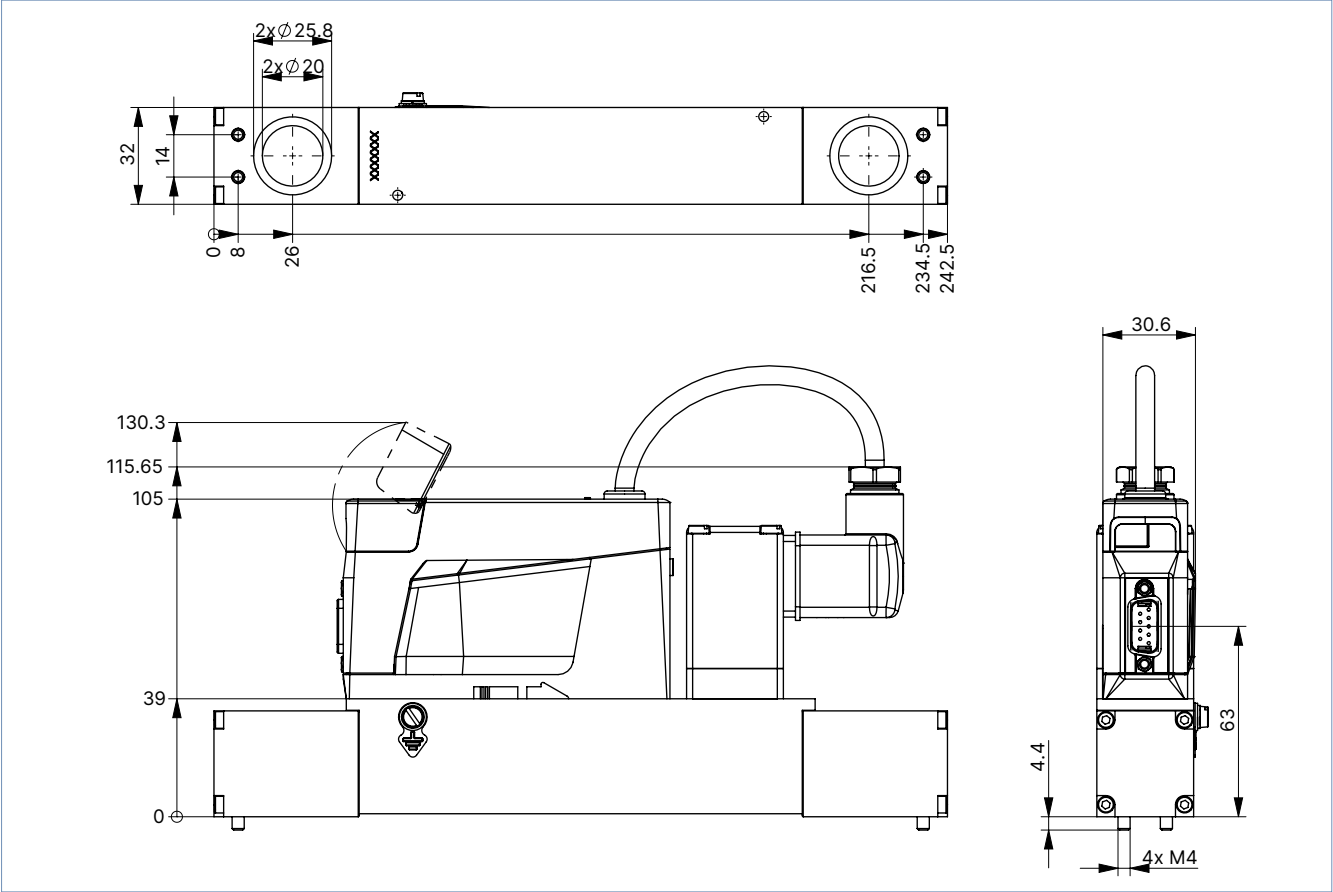
Note:

Dimensions in mm



Sub-base variant

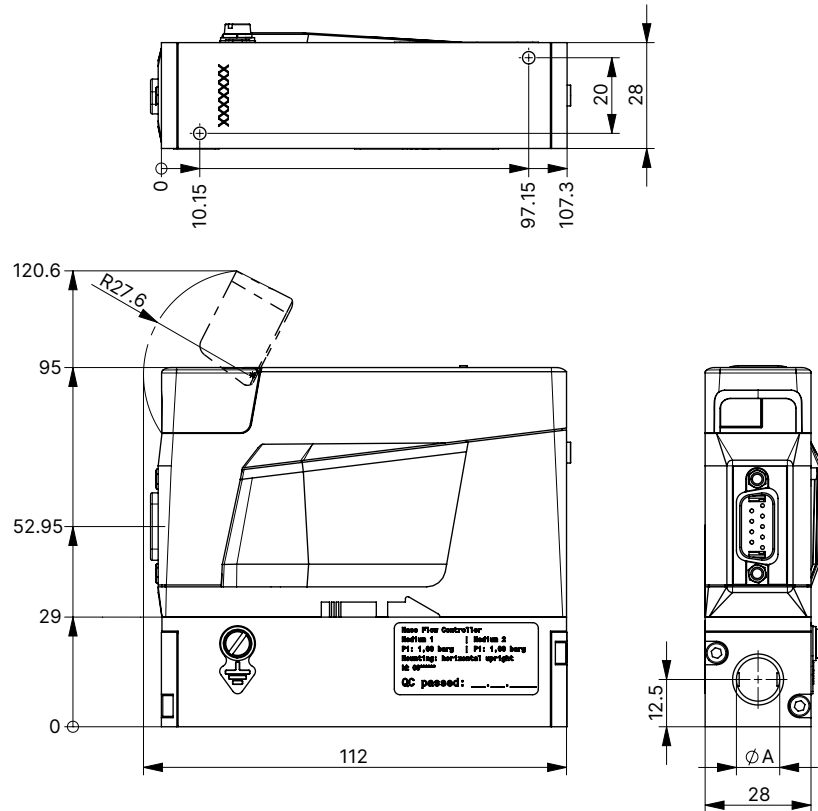
Note:
Dimensions in mm



MFM for nominal flow rates $< 20 \text{ l}_N/\text{min}$

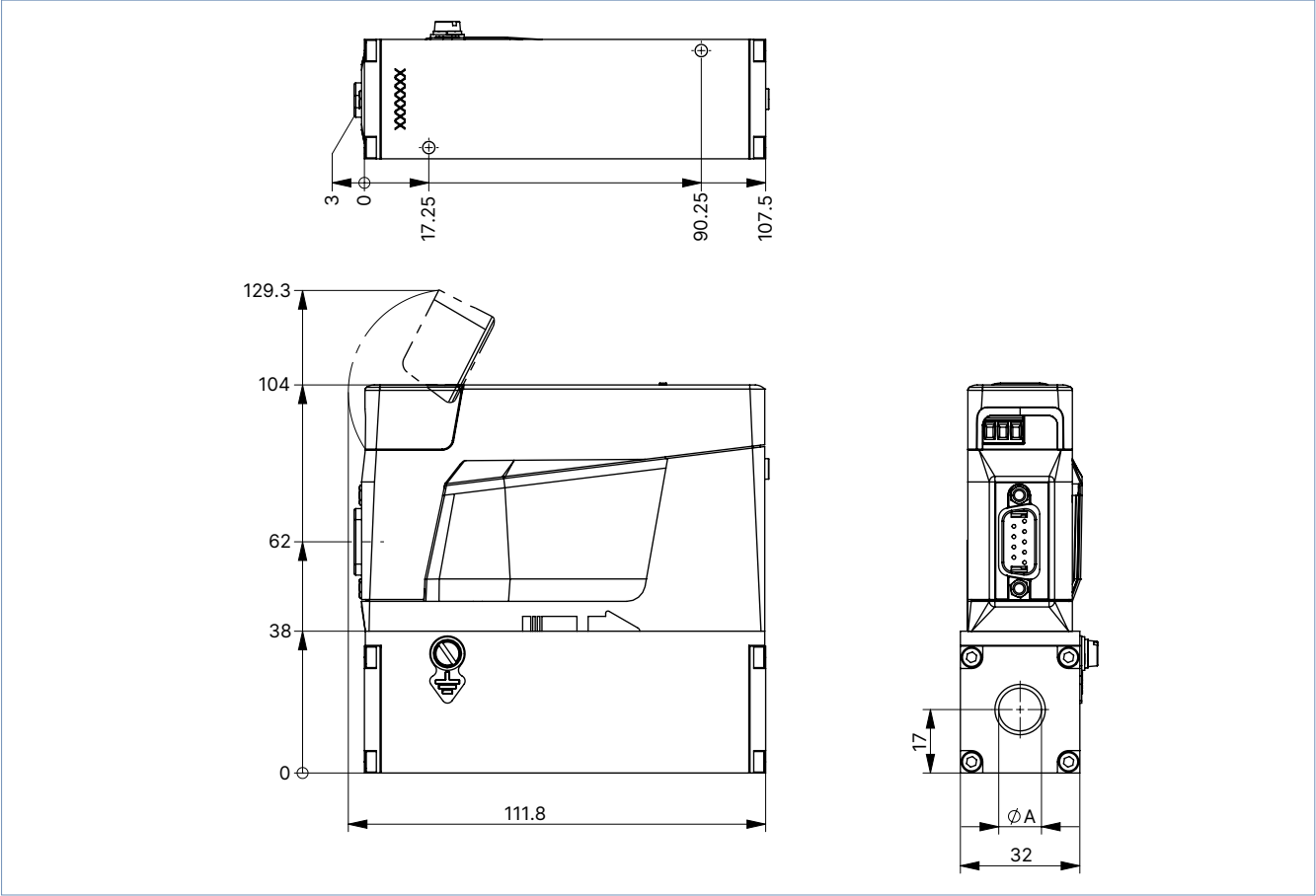
Note:

Dimensions in mm



MFM for nominal flow rates > 20 l_N/min

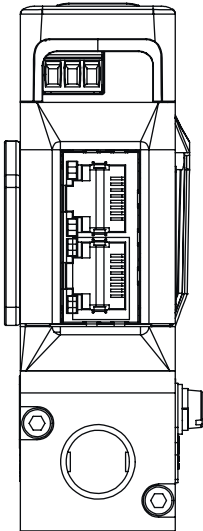
Note:
Dimensions in mm

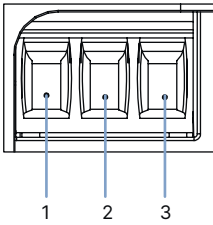


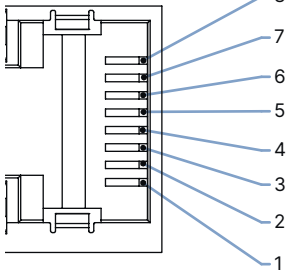
5. Product connections

5.1. Communication

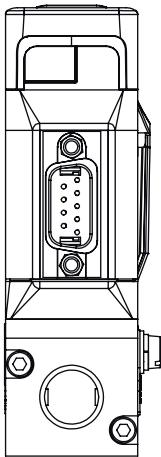
Industrial Ethernet

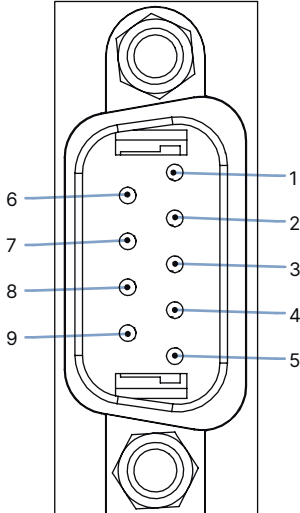


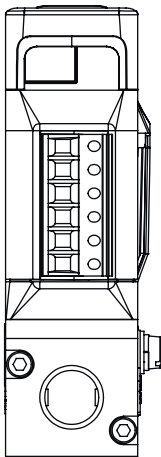
Terminal strip, 3-pin	Pin	Assignment
	1	FE (functional earth)
	2	DGND
	3	24 V

RJ45 socket	Pin	Assignment
	1	TX +
	2	TX –
	3	RX +
	4	Not used
	5	Not used
	6	RX –
	7	Not used
	8	Not used
	Housing	Shielding

Analogue

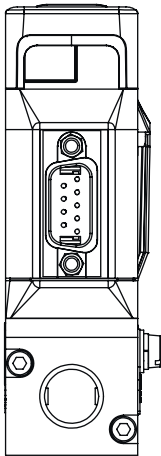


D-Sub plug, 9-pin	Pin	Assignment
	1	Digital input
	2	GND for supply voltage and digital input
	3	24 V
	4	Relay, normally closed contact
	5	Relay, reference contact
	6	Set-point value input +
	7	Set-point value input GND
	8	Actual value output +
	9	Actual value output GND
	Housing	Shielding



Terminal strip, 6-pin	Pin	Assignment
	1	24 V
	2	GND
	3	Set-point value input +
	4	Set-point value input GND
	5	Actual value output +
	6	Actual value output GND

Modbus RTU

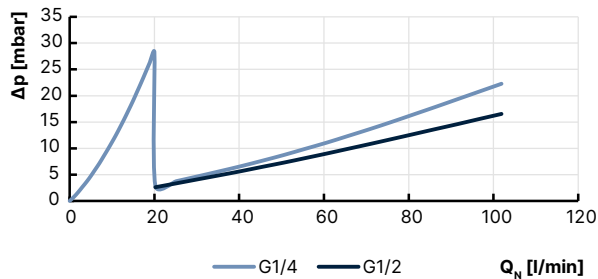


D-Sub plug, 9-pin	Pin	Assignment
	1	Not connected
	2	GND
	3	24 V
	4	Not connected
	5	Not connected
	6	RS-485-Y
	7	RS-485-Z
	8	RS-485-B
	9	RS-485-A
	Housing	Shielding

6. Performance specifications

6.1. MFM pressure loss diagram

The diagram shows an example of the pressure loss curves with nitrogen flowing through. To determine the pressure loss of other gases, the corresponding air equivalent must first be calculated and the base block used for the other gas must be taken into account.



Formula for calculating the pressure loss in an MFM:

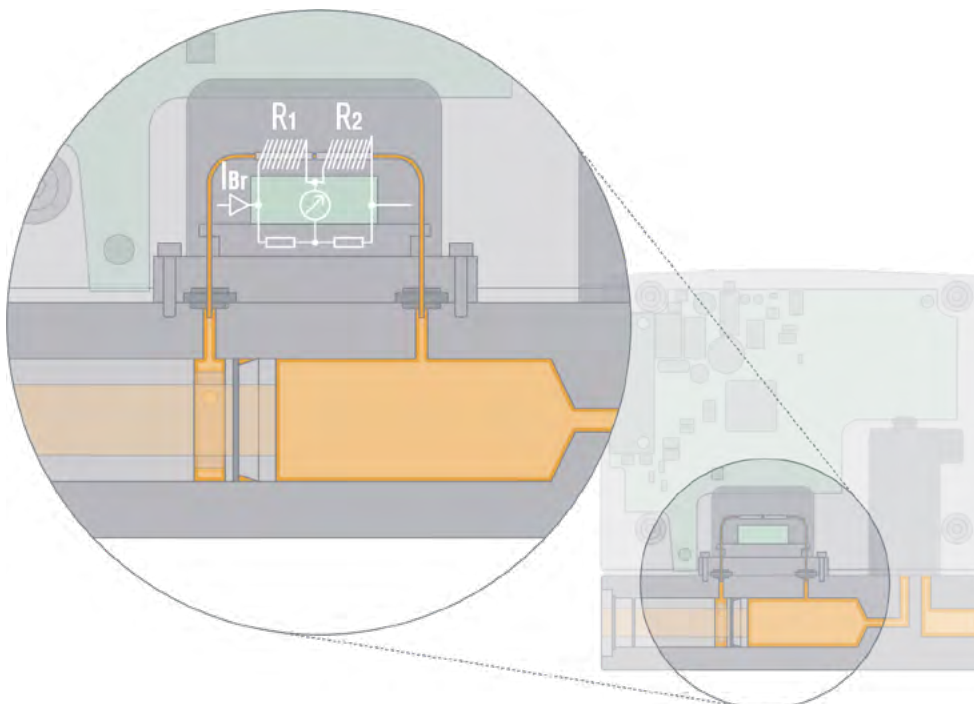
$$\Delta P_{\text{Gas}} = \Delta P_{\text{Air}} \sqrt{\frac{\rho_{\text{N Gas}}}{\rho_{\text{N Air}}}}$$

7. Product operation

7.1. Measurement principle

The measured values are recorded in a secondary channel. A laminar flow element in the main channel creates a slight pressure drop. A part of the gas flow is thereby directed into the secondary channel. This partial flow is proportional to the flow in the main channel (through the laminar flow element). 2 heating resistors are wound around the thin stainless steel tube. With the flow, heat is transported in flow direction. In this way, the heat transport can be used to infer a flow rate value. The sensor tube's wall acts as a thermal barrier and determines measurement dynamics. It is therefore slower than sensors with resistors directly placed in the gas flow.

These sensors can also be used to measure many aggressive gases, as all the wetted main parts are made of stainless steel. This sensor principle also allows conversion between different gases.



8. Product accessories

8.1. Bürkert Communicator software

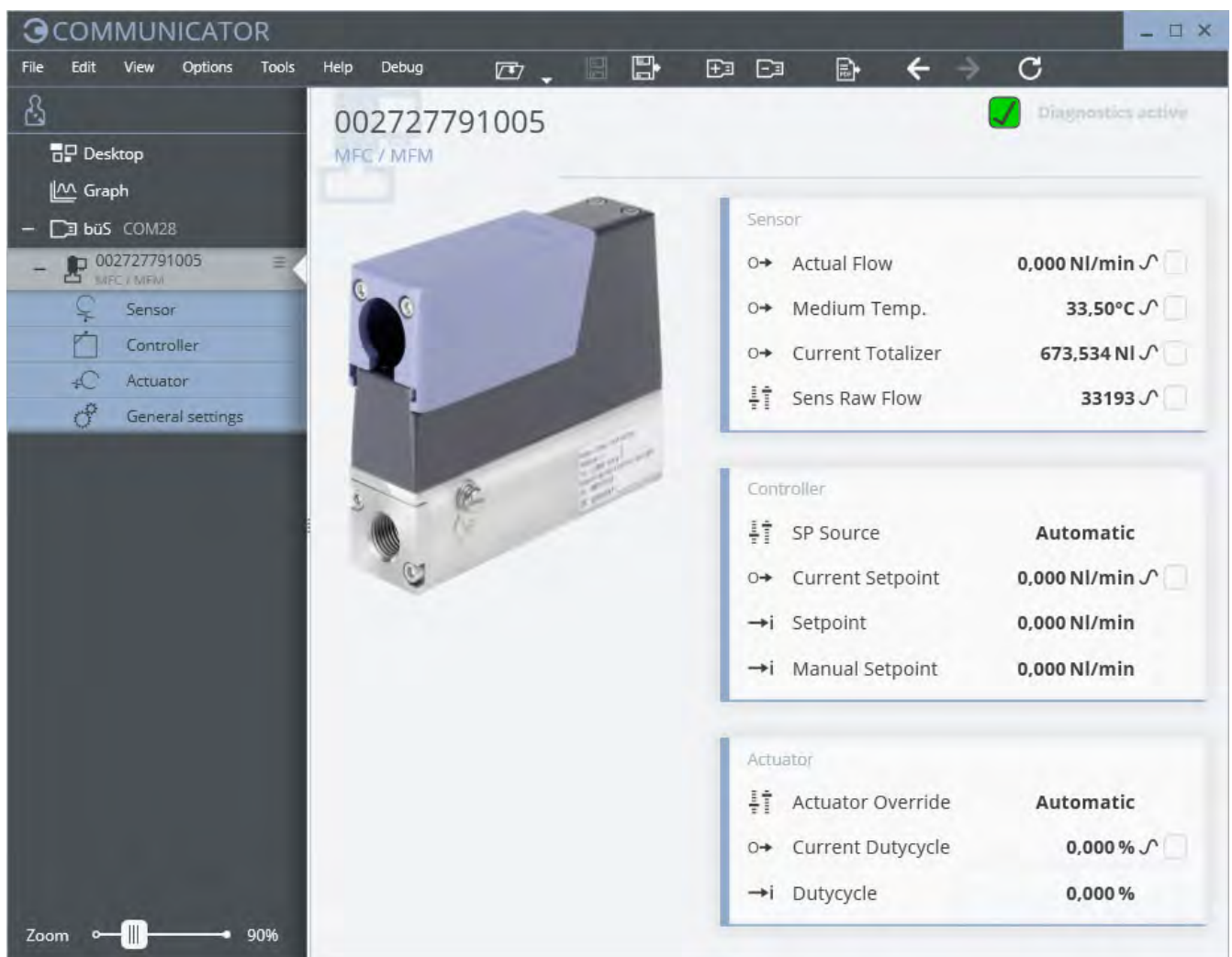
Note:

The corresponding communication software can be downloaded from the website [Type 8920 ►](#).

The Bürkert Communicator is the most important software component of the EDIP (Efficient Device Integration Platform). Various features of this universal tool simplify the configuration and parametrisation of devices equipped with a digital CANopen-based interface. With this tool, the user has a complete overview of cyclic process values as well as acyclic diagnostic data. The integrated graphical programming environment enables the creation of decentralised sub-system control functions. The connection to the PC is established with a USB-büs interface set. The adapter is available as an accessory (see [“9.5. Ordering chart accessories” on page 20](#)).

The Bürkert Communicator enables:

- Configuration, parametrisation and diagnosis of EDIP devices / networks
- Switching between defined gases
- Easy and comfortable mapping of cyclic values
- Graphic display, monitoring and storage of process values
- Firmware update of the connected EDIP devices
- Saving and restoring device configurations
- Zero-point adjustment in case of changed ambient conditions
- Guided re-calibration routine



8.2. Connecting Type 8743 with the Bürkert Communicator software

The Bürkert Communicator interface is based on CANopen. An appropriate bus termination is mandatory. Therefore, activate the switchable terminating resistor on the bÜS stick.

The device is connected via the micro USB socket (USB-bÜS interface set 2 contains the necessary accessories).

Note: no external power supply may be connected to the micro USB socket. The device must be provided with power as described in chapter [“5. Product connections” on page 15](#).

8.3. Configuration management for easy device replacement

If a device needs to be replaced, the memory card can be removed from the defective device and can be inserted into the new device. This transfers all data from the device to be replaced to the new device. The memory card is available as an accessory and must be ordered separately (see [“9.5. Ordering chart accessories” on page 20](#)).

In order to successfully exchange the device, it is necessary that both the new device and the device to be replaced have the same article number.

8.4. Web server for Industrial Ethernet variant

All Industrial Ethernet-based devices (except for the EtherCAT® protocol) from software version A.13.00.00 onwards have an integrated web server. The server can be accessed via a web browser when entering the device's IP address (factory setting IP192.168.1.100).

9. Ordering information

9.1. Bürkert eShop



Bürkert eShop – Easy ordering and quick delivery

You want to find your desired Bürkert product or spare part quickly and order directly? Our online shop is available for you 24/7. Sign up and enjoy all the benefits.

[Order online now](#)

9.2. Recommendation regarding product selection

Note:

Use the Product Enquiry Form (see [“9.4. Bürkert Product Enquiry Form” on page 20](#)) for information on the device layout and send it to us once completed.

For the proper choice of the actuator orifice within the MFC, you need to know not only the required maximum flow rate Q_{N_r} , but also the pressure values directly upstream and downstream of the MFC (p_1 , p_2) at Q_{N_r} . These are often not identical to the inlet and outlet pressure of the entire system, because there are usually additional flow resistances (pipework, additional shut-off valves, nozzles, etc.) both upstream and downstream of the MFC.

Use the Product Enquiry Form to indicate the pressures directly upstream and downstream of the MFC. If the pressure values are unknown or not accessible to measurement, estimates are to be made by taking into account the approximate pressure drops over the flow resistors upstream and downstream of the MFC at Q_{N_r} . The maximum expected inlet pressure $p_{1_{max}}$ must be specified to ensure the tight-closing function of the actuator in all operating states.

9.3. Bürkert product filter

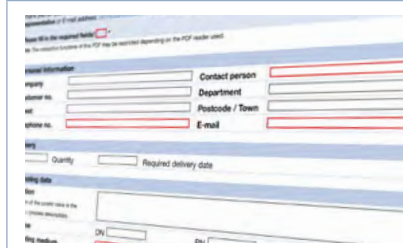


Bürkert product filter – Get quickly to the right product

You want to select products comfortably based on your technical requirements? Use the Bürkert product filter and find suitable articles for your application quickly and easily.

[Try out our product filter](#)

9.4. Bürkert Product Enquiry Form



Bürkert Product Enquiry Form – Your enquiry quickly and compactly

Would you like to make a specific product enquiry based on your technical requirements? Use our Product Enquiry Form for this purpose. There you will find all the relevant information for your Bürkert contact. This will enable us to provide you with the best possible advice.

[Fill out the form now](#)

9.5. Ordering chart accessories

Description	Article number.
General accessories	
USB-büS interface set 2 (Type 8923) for connection to the Bürkert Communicator software: includes büS stick, connection cable to M12 plug, M12 connection cable on micro USB for the büS service interface and Y-distributor, cable length: 0.7 m	772551
Power supply unit Phoenix Class2 (Type 1573), 85...240 V AC/24 V DC, 1.25 A, NEC Class 2 (UL 1310)	772438
Power supply unit for standard rail (Type 1573), 100...240 V AC/24 V DC, 1 A, NEC Class 2 (UL 1310)	772361
Power supply unit for standard rail (Type 1573), 100...240 V AC/24 V DC, 2 A, NEC Class 2 (UL 1310)	772362
Power supply unit for standard rail (Type 1573), 100...240 V AC/24 V DC, 4 A	772363
Memory card	On request
Device description files for software interfaces	Download from Type 8743 ▶
Bürkert Communicator software	Download from Type 8920 ▶
For Type 8743 Analogue	
Terminal strip, 6-pin (standard for Type 8743, included in the scope of delivery of the corresponding analogue variant)	On request
D-sub adapter cable, on strand, cable length: 5 m	580882
D-sub adapter cable, on strand, cable length: 10 m	580883